

IEEE Inter-Society Code Debugging Challenge — SET B Questions

Instructions: Each program below has bugs. Fix them so the program gives the correct output for all test cases. Use the language given and keep the original structure. The bug count is a hint.

Easy = 20 pts · Moderate = 50 pts · Hard = 100 pts · Total = $18 \times 20 + 4 \times 50 + 3 \times 100 = 860$

PART A — EASY (18 × 20 pts)

E1. Sum of Sensor Readings · Python · IEEE CIS · AI&DS · 2 bugs

Concept: Accumulator pattern, loop ranges.

Print the sum of n integers.

Input: n , then n integers **Sample:** 3 / 5 10 20 → 35

```
n = int(input())
a = list(map(int, input().split()))
total = 1
for i in range(1, n):
    total += a[i]
print(total)
```

E2. Peak Voltage · C++ · Signal Processing Society · ECE · 2 bugs

Concept: Finding the maximum, choosing a safe initial value.

Print the largest of n readings (they may be negative).

Input: n , then n integers **Sample:** 3 / -5 -2 -9 → -2

```
#include <bits/stdc++.h>
using namespace std;
int main() {
    int n; cin >> n;
    int mx = 0;
    for (int i = 0; i < n; i++) {
        int x; cin >> x;
        if (x < mx) mx = x;
    }
    cout << mx << endl;
}
```

E3. Factorial · Java · Computer Society · CSE · 2 bugs

Concept: Loop bounds, integer overflow (int vs long).

Print $n!$ ($0 \leq n \leq 20$).

Input: `n` **Sample:** `5` → `120`

```
import java.util.*;
public class Main {
    public static void main(String[] args) {
        int n = new Scanner(System.in).nextInt();
        int f = 1;
        for (int i = 1; i < n; i++) f *= i;
        System.out.println(f);
    }
}
```

E4. Even and Odd Counter · Python · Computer Society · CSE · 1 bug

Concept: The modulo operator, conditions.

Print the number of even values and the number of odd values.

Input: `n`, then `n` integers **Sample:** `5 / 1 2 3 4 6` → `3 2`

```
n = int(input())
a = list(map(int, input().split()))
even = odd = 0
for x in a:
    if x % 2 == 1:
        even += 1
    else:
        odd += 1
print(even, odd)
```

E5. Reverse a Signal Label · C++ · Computer Society · CSE · 2 bugs

Concept: Two-pointer swapping, 0-based indexing.

Reverse a string in place.

Input: one word **Sample:** `hello` → `olleh`

```
#include <bits/stdc++.h>
using namespace std;
int main() {
    string s; cin >> s;
    int n = s.size();
    for (int i = 0; i < n; i++)
        swap(s[i], s[n - i]);
    cout << s << endl;
}
```

E6. Temperature Converter · Java · Circuits & Systems · ECE · 1 bug

Concept: Integer division vs floating-point division.

Convert Celsius to Fahrenheit ($F = C \times 9/5 + 32$) and print it with 1 decimal.

Input: integer C **Sample:** 37 → 98.6

```
import java.util.*;
public class Main {
    public static void main(String[] args) {
        int c = new Scanner(System.in).nextInt();
        double f = c * (9 / 5) + 32;
        System.out.printf("%.1f%n", f);
    }
}
```

E7. Vowel Counter · Python · IEEE CIS · AI&DS · 2 bugs

Concept: String traversal, case handling, += vs ==.

Count the vowels in a line, in either case.

Input: one line **Sample:** IEEE Society → 7

```
s = input()
count = 0
for ch in s:
    if ch in "aeiou":
        count += 1
print(count)
```

E8. Digit Sum · C++ · Computer Society · CSE · 1 bug

Concept: Extracting digits with % 10 and / 10.

Print the sum of the digits of n.

Input: n ($0 \leq n \leq 10^{18}$) **Sample:** 1234 → 10

```
#include <bits/stdc++.h>
using namespace std;
int main() {
    long long n; cin >> n;
    int s = 0;
    while (n > 0) {
        s += n / 10;
        n /= 10;
    }
    cout << s << endl;
}
```

E9. Prime Check · Java · Security · Cyber · 2 bugs

Concept: Trial division up to $\sqrt{n}$, edge cases.

Print PRIME or NOT PRIME.

Input: n ($0 \leq n \leq 10^9$) **Sample:** 7 → PRIME, 1 → NOT PRIME

```
import java.util.*;
public class Main {
    public static void main(String[] args) {
        int n = new Scanner(System.in).nextInt();
        boolean p = true;
        for (int i = 1; (long) i * i <= n; i++)
            if (n % i == 0) p = false;
        System.out.println(p ? "PRIME" : "NOT PRIME");
    }
}
```

E10. Ohm's Law Calculator · Python · Circuits & Systems · ECE/VLSI · 2 bugs

Concept: Type conversion of input, Ohm's law ($I = V/R$, $P = V \times I$).

Given voltage V and resistance R, print current I and power P with 2 decimals.

Input: V R **Sample:** 12 4 → 3.00 36.00

```
V, R = input().split()
I = V / R
P = V * R
print(f"{I:.2f} {P:.2f}")
```

E11. Count the 1-bits · C++ · Circuits & Systems · VLSI · 1 bug

Concept: Binary representation, LSB checking.

Print the number of 1s in the binary form of n.

Input: n ($0 \leq n \leq 10^9$) **Sample:** 13 → 3

```
#include <bits/stdc++.h>
using namespace std;
int main() {
    int n; cin >> n;
    int c = 0;
    while (n > 0) {
        if (n % 2 == 0) c++;
        n /= 2;
    }
    cout << c << endl;
}
```

E12. Average Marks · Java · Computer Society · CSE · 2 bugs

Concept: Array bounds, integer vs double division.

Print the average of n marks with 2 decimals.

Input: n, then n integers **Sample:** 3 / 1 2 2 → 1.67

```
import java.util.*;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int sum = 0;
        for (int i = 0; i <= n; i++) sum += sc.nextInt();
        double avg = sum / n;
        System.out.printf("%.2f%n", avg);
    }
}
```

E13. Linear Search · Python · Computer Society · CSE · 1 bug

Concept: Where to place `return` inside a loop.

Print the 0-based index of the first occurrence of `t`, or -1.

Input: `n t`, then `n` integers **Sample:** `5 7 / 3 9 7 1 7` → `2`

```
n, t = map(int, input().split())
a = list(map(int, input().split()))
def find(a, t):
    for i in range(len(a)):
        if a[i] == t:
            return i
    else:
        return -1
print(find(a, t))
```

E14. GCD of Two Numbers · C++ · Security · Cyber · 1 bug

Concept: Euclid's algorithm, swapping with a temporary variable.

Print `gcd(a, b)`.

Input: `a b` ($1 \leq a, b \leq 10^{18}$) **Sample:** `12 18` → `6`

```
#include <bits/stdc++.h>
using namespace std;
int main() {
    long long a, b; cin >> a >> b;
    while (b != 0) {
        a = b;
        b = a % b;
    }
    cout << a << endl;
}
```

E15. Word Counter · Java · IEEE CIS · AI&DS · 2 bugs

Concept: String splitting with regular expressions, trimming spaces.

Count the words in a line. Words may be separated by several spaces, and the line may have leading or trailing spaces. There is at least one word.

Input: one line **Sample:** `IEEE Computational Intelligence` → `3`

```
import java.util.*;
public class Main {
    public static void main(String[] args) {
        String s = new Scanner(System.in).nextLine();
        String[] w = s.split(" ");
        System.out.println(w.length);
    }
}
```

E16. Atbash Cipher · Python · Security · Cyber · 2 bugs

Concept: Character codes (`ord` / `chr`), substitution ciphers.

Replace each lowercase letter by its mirror ($a \leftrightarrow z$, $b \leftrightarrow y$, ...). Keep spaces unchanged.

Input: lowercase text **Sample:** `ieee cis` → `rvvv xrh`

```
s = input().strip()
res = ""
for c in s:
    if c.isalpha():
        res = chr(ord('z') - ord(c))
    else:
        res += c
print(res)
```

E17. Binary to Decimal · C++ · Circuits & Systems · VLSI · 2 bugs

Concept: Positional number systems, char-to-digit conversion.

Convert a binary string to decimal.

Input: binary string (length ≤ 60) **Sample:** `1010` → `10`

```
#include <bits/stdc++.h>
using namespace std;
int main() {
    string s; cin >> s;
    long long r = 0;
    for (int i = 1; i < (int)s.size(); i++)
        r = r * 2 + s[i];
    cout << r << endl;
}
```

E18. Leap Year · Java · Computer Society · CSE · 1 bug

Concept: Operator precedence of `&&` and `||`, compound conditions.

A year is a leap year if it is divisible by 400, or if it is divisible by 4 but not by 100. Print `LEAP` or `NOT LEAP`.

Input: year **Sample:** `1900` → `NOT LEAP`, `2024` → `LEAP`

```
import java.util.*;
public class Main {
    public static void main(String[] args) {
        int y = new Scanner(System.in).nextInt();
        if (y % 4 == 0 || y % 100 != 0 && y % 400 == 0)
            System.out.println("LEAP");
        else
            System.out.println("NOT LEAP");
    }
}
```

PART B — MODERATE (4 × 50 pts)

BM1. Second Highest Score · Python · Computer Society · CSE · 4 bugs

Concept: Tracking the top two values in one pass, sentinel values.

Print the second largest **distinct** value, or -1 if there isn't one. Values can be negative.

Input: `n`, then `n` integers **Sample:** 5 / 4 9 9 2 7 → 7

```
n = int(input())
a = list(map(int, input().split()))
first = second = 0
for x in a:
    if x > first:
        first = x
        second = first
    elif x > second:
        second = x
print(second)
```

BM2. Bubble Sort · C++ · Computer Society · CSE · 3 bugs

Concept: Bubble sort passes, swapping, early termination.

Sort the array ascending.

Input: `n`, then `n` integers **Sample:** 5 / 5 1 4 2 8 → 1 2 4 5 8

```
#include <bits/stdc++.h>
using namespace std;
int main() {
    int n; cin >> n;
    vector<int> a(n);
    for (auto &v : a) cin >> v;
    for (int i = 0; i < n - 1; i++) {
        bool swapped = false;
        for (int j = 0; j < n - i; j++)
            if (a[j] > a[j + 1]) {
                a[j] = a[j + 1];
                a[j + 1] = a[j];
                swapped = true;
            }
        if (swapped) break;
    }
    for (int i = 0; i < n; i++) cout << a[i] << " \n"[i == n - 1];
}
```

BM3. Anagram Checker · Java · Security · Cyber · 3 bugs

Concept: Frequency counting with arrays.

Print **YES** if the two lowercase words are anagrams, otherwise **NO**.

Input: two words **Sample:** listen silent → YES

```
import java.util.*;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String a = sc.next(), b = sc.next();
        int[] cnt = new int[25];
        for (char c : a.toCharArray()) cnt[c - 'a']++;
        for (char c : b.toCharArray()) cnt[c - 'a']++;
        boolean ok = true;
        for (int x : cnt)
            if (x != 0) ok = false;
            else ok = true;
        System.out.println(ok ? "YES" : "NO");
    }
}
```

BM4. Run-Length Encoding · Python · Signal Processing Society · ECE · 3 bugs

Concept: Data compression, grouping consecutive characters, handling the last group.

Encode each run of characters as the character followed by its count.

Input: one non-empty string **Sample:** aaabccdd → a3b1c2d2

```
s = input().strip()
res = ""
count = 0
for i in range(1, len(s)):
    if s[i] == s[i - 1]:
        count += 1
    else:
        res += s[i - 1] + str(count)
        count = 0
print(res)
```

PART C — HARD (3 × 100 pts)

BH1. Counting Chip Islands (BFS on a Grid) · C++ · Circuits & Systems · VLSI/CSE · 3 bugs

Concept: Breadth-first search, grid traversal with direction arrays, visited marking.

In a grid of 0s and 1s, count the groups of 1s connected up, down, left or right.

Input: r c , then r rows of a 0/1 string ($r, c \leq 1000$)

Sample: 4 5 / 11000 / 11000 / 00100 / 00011 → 3

```
#include <bits/stdc++.h>
using namespace std;
int main() {
    int r, c; cin >> r >> c;
    vector<string> g(r);
    for (auto &s : g) cin >> s;
    vector<vector<bool>> vis(r, vector<bool>(c, false));
    int dx[] = {1, -1, 0, 0}, dy[] = {0, 0, 1, 1};
    int cnt = 0;
    for (int i = 0; i < r; i++)
        for (int j = 0; j < c; j++)
            if (g[i][j] == '1' && !vis[i][j]) {
                cnt++;
                queue<pair<int,int>> q;
                q.push({i, j});
                while (!q.empty()) {
                    auto [x, y] = q.front(); q.pop();
                    vis[x][y] = true;
                    for (int d = 0; d < 4; d++) {
                        int nx = x + dx[d], ny = y + dy[d];
                        if (nx >= 0 && ny >= 0 && nx <= r && ny < c && g[nx][ny] == '1'
&& !vis[nx][ny])
                            q.push({nx, ny});
                    }
                }
            }
    cout << cnt << endl;
}
```

BH2. Longest Unique-Character Key · Java · Security · Cyber · 3 bugs

Concept: Sliding window with two pointers, last-seen index table.

Print the length of the longest substring with no repeated character.

Input: one string of printable ASCII without spaces (length $\leq 10^5$)

Sample: `abcabcbcb` → 3, `abba` → 2

```
import java.util.*;
public class Main {
    public static void main(String[] args) {
        String s = new Scanner(System.in).next();
        int[] last = new int[128];
        int left = 0, best = 0;
        for (int i = 0; i < s.length(); i++) {
            char c = s.charAt(i);
            if (last[c] >= 0) left = last[c] + 1;
            last[c] = i;
            best = Math.max(best, i - left);
        }
        System.out.println(best);
    }
}
```

BH3. Resource Allocation (0/1 Knapsack) · Python · IEEE CIS · AI&DS · 3 bugs

Concept: Dynamic programming, 1-D knapsack with a reverse loop.

Choose items (each at most once) with total weight $\leq W$ and maximum total value.

Input: `n W`, then `n` lines `weight value`

Sample: `3 50 / 10 60 / 20 100 / 30 120` → 220

```
n, W = map(int, input().split())
items = [tuple(map(int, input().split())) for _ in range(n)]
dp = [0] * W
for v, w in items:
    for c in range(w, W + 1):
        dp[c] = max(dp[c], dp[c - w] + v)
print(dp[W])
```